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Longitudinal multi-modal data prediction model for mild cognitive impairment by deep survival analysis

Rong Zhang¹, Durong Chen², Yao Qin², Hongjuan Han³ and Hongmei Yu^{2,4,5*}

Abstract

Background Timely prediction of cognitive decline in patients with Mild Cognitive Impairment (MCI) is crucial for guiding optimal therapeutic interventions. In this study, we aimed to develop a deep survival analysis model that leverages longitudinal, multi-modal data to estimate the probability of dementia conversion, thereby facilitating personalized treatment planning in clinical practice.

Methods We employed a deep neural network model specifically designed for survival analysis to predict the progression from MCI to Alzheimer's Disease (AD). The model integrated longitudinal biomarkers, including neuropsychological assessments and neuroimaging measures, along with baseline demographic characteristics and genetic risk factors, using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database.

Results This study enrolled 922 baseline MCI patients for analysis. The predictive performance was evaluated using a test set at time intervals $\Delta t = 1, 2, 3, 4$ years from the landmark time $s = 1$. The prognostic model exhibited outstanding predictive capability, attaining cAUC values of 0.9089 ± 0.01 alongside BS of 0.1651 ± 0.01 with $\Delta t = 1$ year on the test set, when all variable sets were incorporated into the time-dependent Cox survival neural network (tdCoxSNN) model. Through feature significance evaluation, the Functional Activities Questionnaire (FAQ) emerged as the most influential predictive element.

Conclusions By systematically integrating diverse longitudinal biomarkers, we developed a dynamic prediction model for MCI using deep survival analysis. This approach enables accurate individual risk stratification, facilitates the early identification of high-risk individuals, and supports informed, personalized clinical decision-making.

Keywords Alzheimer's disease, Mild cognitive impairment, Deep survival analysis, Longitudinal multi-modal data, MCI-to-AD

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Background

Alzheimer's disease (AD), recognized as the primary contributor to dementia among elderly populations, represents a gradually worsening neurological condition characterized by memory decline and cognitive dysfunction [1]. In the United States alone, around 7.2 million adults aged 65 and older are currently living with AD dementia, a figure projected to surge to 13.8 million by 2060 without effective disease-modifying interventions [2]. Global dementia cases reached approximately 57 million in 2021 with nearly 10 million new cases annually [3]. Projections indicate this global burden will continue to rise exponentially with population aging, estimated to reach 78 million by 2030 and 152 million by 2050 [4].

Mild cognitive impairment (MCI) has been established as a pivotal intermediary stage between healthy cognition and dementia development, with a well-documented elevated risk of progressing to AD [5]. Meta-analyses and large-scale longitudinal cohort studies (including data from the Alzheimer's Disease Neuroimaging Initiative, ADNI) have consistently reported that the annual conversion rate from MCI to AD ranges from 10 to 15%, translating to approximately 30–45% of MCI cases progressing to AD within 6 years [5, 6]. Epidemiological evidence further supports that MCI serves as a critical prodromal phase for AD, with 40–60% of AD patients having a prior MCI diagnosis [7], highlighting the clinical significance of targeting MCI for early intervention. Early detection of MCI and accurate prediction of its progression trajectory enable healthcare providers to administer timely preventive strategies and personalized interventions, while reliable predictive models can assist in clinical decision-making for treatment planning and identification of high-risk individuals.

A variety of AD-related biomarkers are under investigation to monitor disease progression from MCI to dementia, including cognitive assessments, advanced neuroimaging techniques, and cerebrospinal fluid analyses [8]. Structural Magnetic Resonance Imaging (sMRI) provides insight into cortical atrophy and other structural abnormalities, serving as a critical tool for understanding the mechanisms of neurodegeneration [9]. As the major genetic risk factor, the APOE ϵ 4 allele may contribute to age-related neuronal vulnerability, white matter (WM) damage, and cognitive decline [10–12]. The identification and systematic evaluation of reliable progression markers have the potential to improve diagnostic accuracy for AD and facilitate the development of personalized interventions that optimize clinical outcomes and quality of life for affected individuals [13].

Recent works have applied deep learning models to predict MCI-to-AD conversion using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), yet most approaches solely rely on baseline data such

as a single-timepoint MRI scan, cognitive test scores, or genetic biomarkers, failing to leverage the dynamic, longitudinal data that captures disease progression trajectories over time [14–17]. As a neurodegenerative condition marked by gradual deterioration, AD necessitates longitudinal follow-up to effectively track clinical trajectories in MCI patients and mitigate diagnostic inaccuracies [18]. Many studies have demonstrated that integrating multiple longitudinal biomarkers can substantially enhance predictive performance [19–21]. In contrast, incorporating longitudinal data—including serial MRI scans, repeated neurocognitive assessments, and sequential clinical measurements—offers notable advantages: it enables the capture of subtle, progressive changes in brain structure and cognitive function that are often undetectable in cross-sectional baseline data, thereby improving the model's distinguish ability [18]. Many studies have demonstrated that integrating multiple longitudinal biomarkers can substantially enhance predictive performance [19–21].

Most current deep learning methods have been primarily applied to the classification and diagnosis of AD [22–24], while their use in AD prognosis or prediction remains relatively limited—especially studies leveraging longitudinal data [25–27]. Relevant deep learning-based works focused on AD prediction are summarized in Table 1. Given the critical role of incorporating multiple biomarkers and implementing longitudinal follow-up, we developed a dynamic prediction model for individual risk prediction, building on and extending prior deep survival research. The model combined time-dependent Cox model [28] with a deep neural network to represent the nonlinear relationship between longitudinal processes and time-to-event processes. Notably, the methodology eliminates dependence on restrictive prior assumptions or conventional parametric constraints while maintaining predictive accuracy, addressing limitations of some earlier models. Second, it effectively leverages existing longitudinal data for model training without requiring explicit modeling of longitudinal trajectories, offering advantages in handling complex temporal dynamics compared to discrete-time approaches. Additionally, we conduct a systematic comparison of the predictive performance of 3 distinct multi-modal biomarker combinations (encompassing demographic characteristics, neuroimaging, neuropsychological assessments, genetic), providing actionable insights into optimal biomarker selection for clinical application.

Methods

Participants

All data in this study were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database, a multicenter, longitudinal observational study designed to

Table 1 Major studies on deep learning methods for AD prediction

Ref	Model	Dataset used Modalities	Results	Limitations
[14] (2023)	Deepsurv	ADNI Genetic (SNP, APOE) MRI CDC(Cognitive test, Demographic, CSF)	C-index: 0.831 (MRI+CDC) IBS: 0.105 (MRI+CDC)	1. This study solely employed baseline data rather than longitudinal data. 2. The MRI-based neuroimaging features used in this study were derived from the FreeSurfer segmented structural volumes.
[15] (2023)	S-CNN MLP, GMV Weibull CPH MLP, GMV+CSF	ADNI, NACC MRI CSF (A β , tau) MMSE Demographic	C-index: IBS: 0.756 0.0209 (0.051) (0.050) 0.0731 0.192 (0.032) (0.057) 0.743 0.205 (0.020) (0.049) 0.750 0.182 (0.051) (0.055) 0.729 0.181 (0.022) (0.044)	1. No other imaging data such as tau PET or functional MRI. 2. Only baseline data was used, without using longitudinal data. 3. Not considering the reversal from MCI to pre-MCI state. 4. Based on the pre-segmented brain region volumes without directly using the original images.
[16] (2025)	SRNet+MRNet	ADNI, The China Longitudinal Aging Study (CLAS) T1-weighted MRI Demographic Cognitive test	AUC: 0.812	1. Only baseline data is available, with no follow-up data. 2. Only using T1-weighted MRI, without integrating other T1-weighted MRI. 3. The research only included two cohorts and did not verify the data through collaboration with other research institutions.
[17] (2025)	Fast RSF	ADNI, AIBL Demographic features Cognitive score features Comorbidity features	C-index: 0.8607	1. This study is limited by the use of only baseline features, without accounting for longitudinal changes in comorbidities or cognitive scores. 2. To address dataset mismatch concerns, the study retrained the fast random survival forest model on ADNI using only the common features shared with AIBL.
[25] (2025)	AutoGluon	ADNI Demographic Genetic data Neuropsychological assessment Plasma and CSF biomarker measurements sMRI	AUC: 0.87–0.92	1. The study lacks an external validation cohort. 2. Due to the relatively small sample size, the study did not perform subgroup analysis.
[26] (2025)	RestNet3D T-LSTM	ADNI T1-weighted MRI	C-index: 0.80–0.91	1. Limited sample size of this study restricted secondary analyses. 2. The study did not investigate more complex CNN architectures that were shown to have demonstrated higher performance on traditional benchmarks. 3. Sociopolitical constraints limited our ability to validate findings externally.
[27] (2025)	Modified DenseNet-BC 3D Grad-CAM	ADNI T1-weighted 3D MRI Cognitive test scores	AUC: 0.9191 $\pm$ 0.0560 BA: 0.8347 $\pm$ 0.0261	1. The sample size of MRI data was small result in overfitting. 2. The study only explored a single type of CNN architecture and evaluated one way of creating a feature extractor model. 3. The study used only anatomical MRI data.

integrate comprehensive neuroimaging, clinical, and neuropsychological assessment data for tracking the progression of Alzheimer's disease. Written informed consent was obtained from all participants. Detailed information regarding the study protocols, including inclusion and exclusion criteria, is available at <http://www.adni-info.org/>. For the ADNI study, written informed consent was obtained for all participants and the study protocol was approved by the institutional review board at each participating center before protocol-specific procedures were performed. We constructed a retrospective cohort by reviewing subjects from 2005 through December 2022.

Participants were included if they (1) were diagnosed with MCI at study entry and (2) underwent at least two visits during the study period until AD diagnosis or until December 2022, whichever came first. Cognitive status was systematically reassessed approximately every six months or annually, and was classified as either MCI or AD. Detailed diagnostic criteria for assessing cognitive status are provided in the Supplementary Material.

Variables

The predictors consisted of baseline characteristics and longitudinal variables. Baseline characteristics

encompassed the demographic information including educational attainment, gender, age range, and marital status, along with genetic predisposition indicators such as apolipoprotein E (ApoEε4) allele carriage status. Longitudinal variables from two modalities of biomarkers i.e. cognitive function and structural MRI morphometry were also analyzed. Cognitive functioning underwent comprehensive assessment using eleven neuropsychological assessments: Alzheimer's Disease Assessment Scale (ADAS), Functional Activities Questionnaire (FAQ), Rey Auditory Verbal Learning Test (RAVLT), Trial Making Test-version B (TMT-B), modified Preclinical Alzheimer's Cognitive Composites with Digit Symbol Substitution (mPACCdigit), modified Preclinical Alzheimer's Cognitive Composite with Trail B (mPACCtrailsB). Detailed descriptions and calculation methods regarding cognitive assessments are available in the Supplementary Material. It should be noted that in the diagnostic criteria for MCI and AD in the ADNI database, MMSE (Mini-mental state evaluation) and CDRSB (Clinical dementia rating sum of boxes) were all taken into account, which were excluded in the subsequent construction of all prediction models in this study. We applied MICE (Multiple Imputation by Chained Equations, MICE) for missing data. MICE is a widely used and robust technique that generates multiple plausible values for each missing data point based on the observed data, thereby reducing potential biases associated with missing data and improving statistical power. MICE is well-suited to preserve the relationships between variables and avoid the potential biases that might arise from simpler methods such as mean imputation or listwise deletion.

Structural MRI (sMRI) data underwent spatial normalization through projection onto a comprehensive gray matter density framework, incorporating 116 neuroanatomical divisions as per the automated anatomical labeling protocol. This parcellation scheme differentiated cerebral territories into 90 cortical subcortical zones and 26 cerebellar subdivisions [29]. Due to the long time span of the ADNI study, variations in MRI equipment manufacturers, system configurations, and multicenter data acquisition, sequence parameters of neuroimaging may differ. To enhance the availability and mitigate the impact of interscanner variability, a normalization protocol was implemented involving determination of proportional gray matter density values for individual brain structures relative to total intracranial volume. Additional details regarding the preprocessing and feature extraction procedures for sMRI are provided in the Supplementary Material.

Model construction

Let $D_n = \{T_i, \delta_i, X_i, Y_i(T_i); i = 1, \dots, n\}$ represent the observed data from n independent subjects. For the i -th

subject, the observed time T_i is defined as $\min(T_i^*, C_i^*)$, where T_i^* is the true event time and C_i^* is the censoring time, and $\delta_i = I(T_i^* \leq C_i^*)$ is event indicator. X_i denotes a vector of baseline (time -invariant) covariates. We let $y(t)$ denote the longitudinal measurement taken at time point t . $Y_i(t) = \{y_i(t_{il}) | 0 \leq t_{il} \leq t, l = 1, \dots, n_i\}$ are the n_i longitudinal measurements for subject i between time 0 and t .

Dynamic prediction focuses on estimating the survival probability of a new patient j beyond a future time point $u(u > s)$, given their time-varying measurements up to time s . This probability, denoted as $\pi_j(u|s) = \Pr(T_j^* > u | T_j^* > s, Y_j(s), X_j)$, is conditional on the patient having survived until s , along with their observed history and baseline covariates. Unlike static approaches, dynamic models enable the revision of forecasts as new data become available; for instance, an updated prediction $\hat{\pi}_j(u|s')$ can be obtained when additional information is recorded at a later time $s' > s$.

The time-dependent Cox model assumes:

$$h_i(t|X_i, Y_i(t)) = h_0(t) \exp(\beta^T X_i + \gamma^T y_i(t)) \quad (1)$$

Where $h_0(t)$ is the baseline hazard function and $h_i(t|X_i, Y_i(t))$ is the hazard function of the i th individual with covariates X_i and $Y_i(t) = \{y_i(t'), 0 \leq t' \leq t\}$. The coefficients (β, γ) are estimated through the partial likelihood

$$\prod_i^n \left\{ \frac{\exp[g_{\beta, \gamma}(X_i, y_i(T_i))]}{\sum_{j: T_j \geq T_i} \{\exp[g_{\beta, \gamma}(X_j, y_j(T_i))] - E_j(g_{\beta, \gamma})\}} \right\}^{\delta_i}$$

Where $g_{\beta, \gamma}(X_i, y_i(T_i)) = \beta^T X_i + \gamma^T y_i(T_i)$ is the risk score function.

$$E_j(g) = \frac{\sum_k 1(k > j, T_k = T_j)}{\sum_k 1(T_k = T_j)} \sum_{k: T_k = T_j} \exp[g(X_k, y_k(T_j))]$$

is Efron's approximation for handling tied events [30]. In this work, we use the instantaneous measurement at time t (i.e., $y_i(t)$) as the time-dependent predictor in the model (1).

The time-dependent Cox survival neural network (tdCoxSNN) [31] expand the time-dependent Cox model using a neural network to replace the linear effect of the covariates in the model (1). Specifically, tdCoxSNN is a feed-forward neural network that models the nonlinear effect of time-dependent covariates on the hazard function. Instead of assuming the risk score function $g(X_i, y_i(t))$ takes the linear form, we leave the form of $g(X_i, y_i(t))$ unspecified and model it through a neural network $g_\theta(X_i, y_i(t))$, parameterized by $\theta = (W, V)$, where W is the weight matrices and V is the shift vectors

of the neural network. The input of the neural network is the predictors $X_i, y_i(t)$ and the output is a single node that represents the risk score $g_\theta(X_i, y_i(t))$. The parameter θ is estimated by minimizing the negative log partial likelihood $\hat{\theta} = \arg \min_{\theta} l(\theta)$, where

$$l(\theta) = -\frac{1}{n} \sum_{i=1}^n \delta_i \left[g_\theta(X_i, y_i(T_i)) - \log \left\{ \sum_{j: T_j \geq T_i} \exp(g_\theta(X_j, y_j(T_i))) - E_i(g_\theta) \right\} \right] \quad (2)$$

The tdCoxSNN model development utilized Python (version 3.12) and the models were developed using PyTorch (version 2.1.0). In the analysis of real data, we employed survival neural network architecture structured as follows: input layer $\rightarrow$ one hidden layer $\rightarrow$ batch normalization layer $\rightarrow$ dropout layer $\rightarrow$ output layer. The batch normalization layer [32] enhances training speed by stabilizing the learning process, while the dropout layer [33] helps prevent overfitting during model training. The same set of hyper-parameters is consistently applied across all analyses: a hidden layer with 30 neurons, the Scaled Exponential Linear Unit (SeLU) as the activation function for the hidden layer, a batch size of 50, 20 training epochs, a learning rate of 0.01, and a dropout rate of 0.2. Optimization of the neural network is performed using the Adam algorithm [34].

Multi-modal survival comparison

Multi-modal data fusion integrates diverse data sources to provide a comprehensive characterization of disease profiles and facilitate the identification of AD-related biomarkers. To evaluate the prognostic utility of different biomarker combinations for predicting MCI-to-AD conversion, we developed predictive models incorporating four demographic variables, ApoE ϵ 4 status, neuropsychological assessments, and neuroimaging data. Three distinct feature sets were constructed, while maintaining consistent hyperparameter configurations across all models:

FS1: Demographic information (4 features)+ ApoE ϵ 4 + sMRI data (116 features)

FS2: Demographic information (4 features) + ApoE ϵ 4 + Neuropsychological assessments (11 features)

FS3: Demographic information (4 features) + ApoE ϵ 4 + Neuropsychological assessments (11 features) + sMRI data (116 features)

To assess the predictive performance of three features sets, we employed three distinct models: the time-dependent Cox model (tdCox), time-dependent Cox survival neural network (tdCoxSNN) [28] and the discrete-time deep learning framework Dynamic-DeepHit [35].

The dataset was randomly split into a training set and an independent test set (8:2 ratio, $n = 184$ for the test set). Ten-fold cross-validation (10-fold CV) was performed on the training set for model development: nine subsets were used for training in each iteration, and one subset for internal validation. The mean $\pm$ standard deviation values presented in the results were obtained from the independent test set. The cumulative dynamic AUC (cdAUC) and Brier Score (BS) were used to assess the performance of the predictive models. Higher scores of cdAUC and lower scores of BS indicate superior performance, respectively.

To evaluate feature significance, measurements of the same features across multiple time points were grouped into unified categories. This study utilized the sparse group lasso approach [36], integrating both lasso and group lasso penalty terms, to assess the significance of features. Through structured penalties, the group lasso not only can screen out important features, but also can reveal the overall importance pattern of the features over time. Specifically, it preserves non-zero groups while simultaneously selecting key elements within each group, achieving both inter-group sparsity and intra-group sparsity.

Evaluation metrics

The evaluation of predictive performance in survival models primarily focuses on two key dimensions: calibration and discrimination. Calibration measures the agreement between the predicted probabilities and the observed outcomes, while discrimination evaluates a model's ability to distinguish between subjects who experience the event and those who do not. Both types of metrics were incorporated in assessing the performance of our models. Consistent with previous studies on joint modeling and landmarking approaches, dynamic prediction models were compared within a predefined time interval $(s, s + \Delta t]$, where s denotes the landmark time point [37, 38].

Time-dependent AUC

The area under the receiver operating characteristic curve (AUC) is used to evaluate the discriminative performance of model, with values ranging from 0 to 1. A value closer to 1 indicates stronger discrimination ability. In this study, we use cumulative dynamic AUC (cdAUC) [39] is used to assess the model's discriminatory performance across different time points. Let T_i denotes the time of disease onset and X_i is a marker value (usually the value at baseline) for individual i ($i = 1, 2, \dots, n$). For a given threshold c , the cumulative sensitivity is defined as $Se^C(b, \Delta t) = P(X_i > b | s < T_i \leq s + \Delta t)$ and the dynamic specificity is defined as $Sp^D(b, \Delta t) = P(X_i \leq b | T_i > s + \Delta t)$. With the

cumulative sensitivity and dynamic specificity, the corresponding $\text{cdAUC}(s, \Delta t)$ is defined as

$$\text{cdAUC}(s, \Delta t) = P(X_i > X_j | s < T_i \leq s + \Delta t, T_j > s + \Delta t, i \neq j). \quad (3)$$

Time-dependent Brier Score

The time-dependent Brier Score assesses model calibration by computing the mean squared error between observed survival status and predicted survival probabilities, incorporating inverse probability of censoring weighting (IPCW) [40] to account for censored observations. The score ranges from 0 to 1, with scores closer to 0 denoting excellent calibration. For a given landmark time s , the estimated Brier Score at time $s + \Delta t$ is

$$\hat{BS}(s, \Delta t; \hat{\pi}) = \frac{1}{\sum_i 1(T_i > s)} \sum_{i=1}^n \left[1(T_i > s) \hat{W}_i(s, \Delta t) \left\{ 1(T_i > s + \Delta t) - \hat{\pi}_i(s + \Delta t | s) \right\}^2 \right] \quad (4)$$

Where $\hat{W}_i(s, \Delta t) = \frac{1(T_i > s + \Delta t)}{\hat{G}(s + \Delta t | s)} + \frac{1(T_i \leq s + \Delta t) \delta_i}{\hat{G}(T_i^- | s)}$ is the IPCW weight and $\hat{G}(u | s) = \frac{\hat{G}(u)}{\hat{G}(s)}$ is the Kaplan-Meier estimate of the conditional censoring distribution $Pr(C^* > u | C^* > s)$ with $u > s$.

Results

To further evaluate the performance of tdCoxSNN, we compared it with the tdCox model and the Dynamic-DeepHit model across three feature sets (FS1, FS2, and FS3). In all experiments, MCI patients were temporally aligned based on the date of their first follow-up visit to ensure consistent comparison of risk predictions across time. Specifically, the timing of both measurements and event or censoring occurrences was defined as the time elapsed since the first follow-up visit. Given that the follow-up period spans six months, we discretized time with a resolution of six months.

Characteristics of participants

The research workflow is illustrated in Fig. 1. This study included 4060 observations from 922 subjects with MCI, with 738 (80%) cases allocated to the training set and 184 (20%) to the test set. The median follow-up time was 2 years (interquartile range, 1–4 years), with total follow-up ranging from 0.42 to 14.67 years. In the training set, 283 cases (38%) progressed from MCI to AD, while 455 cases (62%) remained in the MCI state. In the test set, 74 cases (40%) progressed to AD, and 110 cases (60%) remained in the MCI state. Descriptive statistics for baseline variables are presented in Table 2, and more detailed

information on the training and test sets is provided in Table 3.

Performance of predictive model

For comparative evaluation, we fitted the tdCox, Dynamic-DeepHit and tdCoxSNN models using three feature sets. Predictive performance was assessed on a test set of 184 cases at time intervals $\Delta t = 1, 2, 3, 4$ years from the landmark time $s = 1$. Figure 2 illustrates box-plots of cdAUC and BS for the three methods across the feature sets FS1, FS2, and FS3. The results indicate that the tdCoxSNN model outperformed the other two methods in both evaluation metrics. Notably, tdCoxSNN achieved its highest performance on FS3, highlighting the critical importance of incorporating brain structural measurements and demonstrating the model's ability to leverage neuroimaging-clinical complementarity. Conversely, its performance was lowest on FS1, likely due to the absence of neuropsychological assessment features, underscoring their essential role in predicting disease progression with tdCoxSNN.

For FS1, Dynamic-DeepHit and tdCox showed comparable predictive accuracy in terms of cdAUC, while Dynamic-DeepHit achieved the lowest calibration error (BS). On FS2, Dynamic-DeepHit attained the highest cdAUC, whereas tdCoxSNN achieved the lowest Brier Score. For FS3, tdCox and tdCoxSNN exhibited similar BS, while Dynamic-DeepHit showed the lowest cdAUC performance. Interestingly, integrating all variables into the Dynamic-DeepHit model did not improve performance and, in some cases, negatively impacted it. Across all models, predictive performance declined as the prediction horizon increased. This reflects the dynamic nature of longitudinal predictors and limited future information, rather than shortcomings of the modeling approach. Meanwhile, The predictions are conditional on survival up to the landmark time and that uncertainty increases for longer prediction horizons.

Feature importance analysis

To evaluate the importance of individual features in our model, we employed the average absolute coefficient value across K-fold cross-validation. This method provides a stable, global estimate of each feature's contribution to the model's predictions. The results of the group feature importance analysis on models trained with all features is shown in Fig. 3(a). Neuropsychological assessments dominated the highest-ranked attributes, with the Functional Activities Questionnaire (FAQ) emerging as the paramount predictor, succeeded by mPACCtrailsB in subsequent ranking. The fifth important characteristic was the neuroimaging feature R56, which is located in the right fusiform gyrus (FFG). Figure 3(b) illustrated the

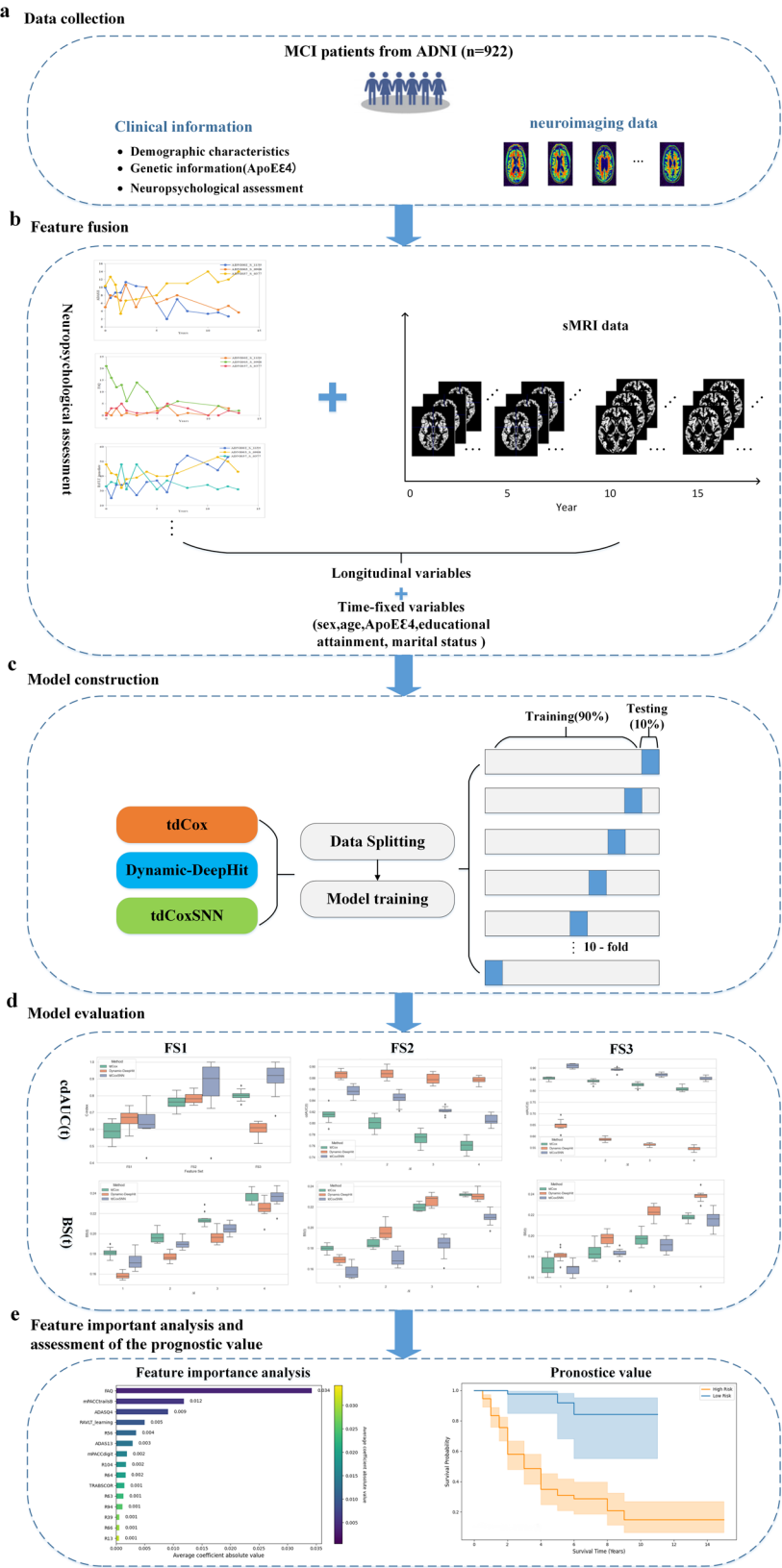


Fig. 1 Flow diagram of the study. (a-e) sequentially integrate data collection, three feature fusion strategies, model development and evaluation for tdCox, dynamic-DeepHit, tdCoxsnn, as well as for prognostic value assessment

Table 2 The baseline variables of the progression of MCI individuals

Variables	Training set (n = 738)		P value	Test set (n = 184)		P value
	Non-progress to AD (n = 455)	Progress to AD (n = 283)		Non-progress to AD (n = 110)	Progress to AD (n = 74)	
Age (years)	74.54 ± 7.77	74.72 ± 7.02	0.754	75.14 ± 7.44	73.96 ± 7.06	0.280
Sex			0.790			0.748
Male	182 (40.4)	116 (41.0)		42 (38.2)	30 (40.5)	
Female	273 (59.6)	167 (59.0)		68 (61.8)	44 (59.5)	
Educational attainment (years)	15.97 ± 2.85	15.96 ± 2.70	0.969	15.72 ± 2.86	15.69 ± 2.82	0.946
Marital Status			0.302			0.175
Married	352 (77.4)	228 (80.1)		83 (75.5)	62 (83.8)	
Single	103 (22.6)	55 (19.9)		27 (24.5)	12 (16.2)	
ApoEε4			<0.001			0.044
0	262 (57.6)	104 (36.7)		65 (59.1)	30 (40.5)	
1	156 (34.3)	140 (47.5)		31 (28.2)	32 (43.2)	
2	37 (8.1)	39 (13.8)		14 (12.7)	12 (16.3)	

Table 3 The baseline variables of training set and test set

Variables	Training set (n = 738)	Testing set (n = 184)	Total	P value
Age (years)	74.61 ± 7.45	74.67 ± 7.30	74.62 ± 7.45	0.926
Sex				0.757
Male	298 (40.4)	72 (39.1)	370 (40.1)	
Female	440 (59.6)	112 (60.9)	552 (59.9)	
Educational attainment (years)	15.97 ± 2.80	15.71 ± 2.84	15.91 ± 2.80	0.261
Marital Status				0.95
Married	580 (78.6)	145 (78.8)	725 (78.6)	
Single	158 (21.4)	39 (21.2)	197 (21.4)	
ApoEε4				0.184
0	366 (49.6)	95 (51.6)	461 (50.0)	
1	296 (40.1)	63 (34.2)	359 (38.9)	
2	76 (10.3)	26 (14.2)	102 (11.1)	

top 30 features within the feature set FS3 across various temporal intervals.

Neuroimaging features accounted for 30% of the significant variables, whereas neuropsychological assessment features comprised the remaining 70%. Among all variables, the FAQ (Functional Activities Questionnaire, FAQ) measured at the second-year follow-up emerged as the most important predictor. The most influential neuroimaging feature was R56_time 3.0, which represents the percentage of gray matter density in the occipital lobe, specifically within the fusiform gyrus, measured in the third year relative to the entire brain tissue.

Prognostic risk stratification assessment

Based on the estimated risk score $\hat{g}_\theta(X_i, y_i(t))$ produced by tdCoxSNN on the training set, the test set was uniformly divided into high-risk and low-risk groups.

The Log-rank test demonstrated that overall survival (OS) in the test set could be significantly stratified according to the output estimated risk score $\hat{g}_\theta(X_i, y_i(t))$.

Subsequently, a Gaussian mixture model was fitted to the test set risk scores, identifying two distinct subgroups, as illustrated in Fig. 4. Comparative analysis of Kaplan-Meier survival curves revealed marked differences in disease progression between these subgroups. Over the ten-year observation period, 55% of individuals in the high-risk group progressed to AD, compared with fewer than 4% of participants in the low-risk group, as shown in Fig. 5.

Discussion

In this study, we employed a deep neural network model specifically designed for survival analysis with longitudinal biomarkers to predict the progression from MCI to AD. The tdCoxSNN model demonstrated effective individualized risk prediction and has the potential to facilitate early identification of high-risk individuals, thereby supporting clinical decision-making. Incorporating demographic information, ApoEε4 status, neuropsychological assessments, and neuroimaging data significantly improved predictive performance. Furthermore, we assessed the relative importance of different biomarkers, identifying the FAQ as the most important neuropsychological feature and the FFG as the most critical neuroimaging feature.

Dynamic prediction [41] leverages longitudinal variables collected during follow-up to continuously update risk estimates based on patients' most recent data. Current approaches for dynamic prediction include joint modeling, which often suffers from computational complexity and reduced performance under model misspecification, and landmarking, which is straightforward to implement but is typically constrained by reliance on proportional hazards assumptions [42, 43]. Recent advances in artificial intelligence have enabled the development of novel survival analysis frameworks using deep neural networks [44–47], effectively overcoming

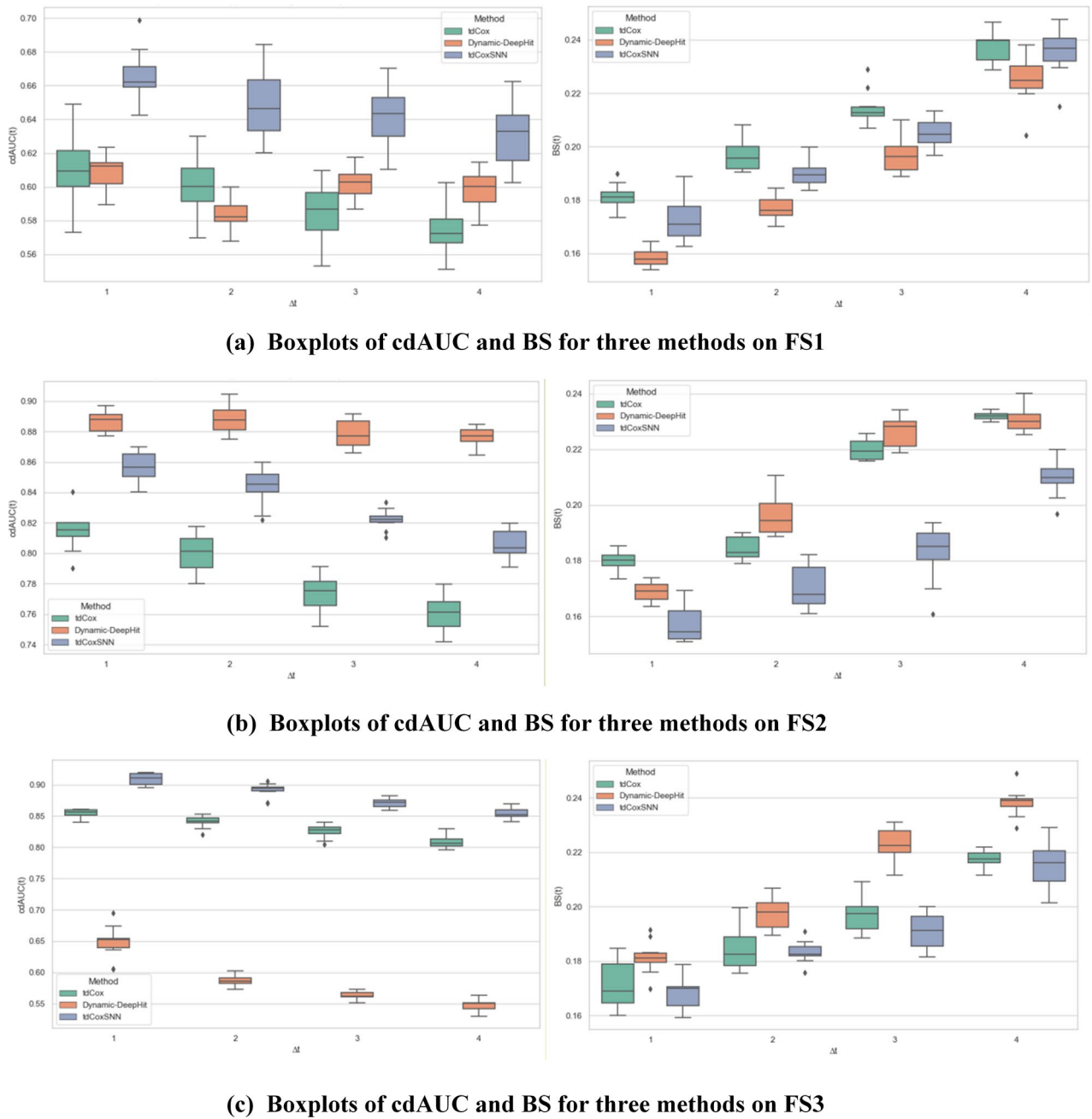
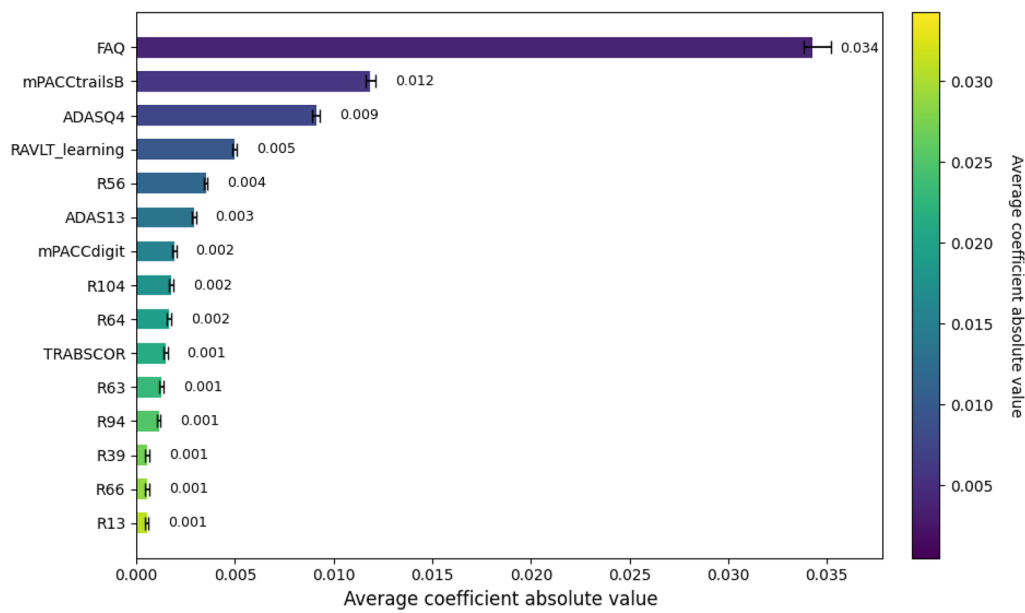


Fig. 2 Boxplots of cdAUC and Bs at $\Delta t = 1, 2, 3, 4$ from landmark time $s = 1$ for three methods on the three feature sets. Methods included in the comparison are the tdCox, the dynamic-DeepHit, and the tdCoxsnn model

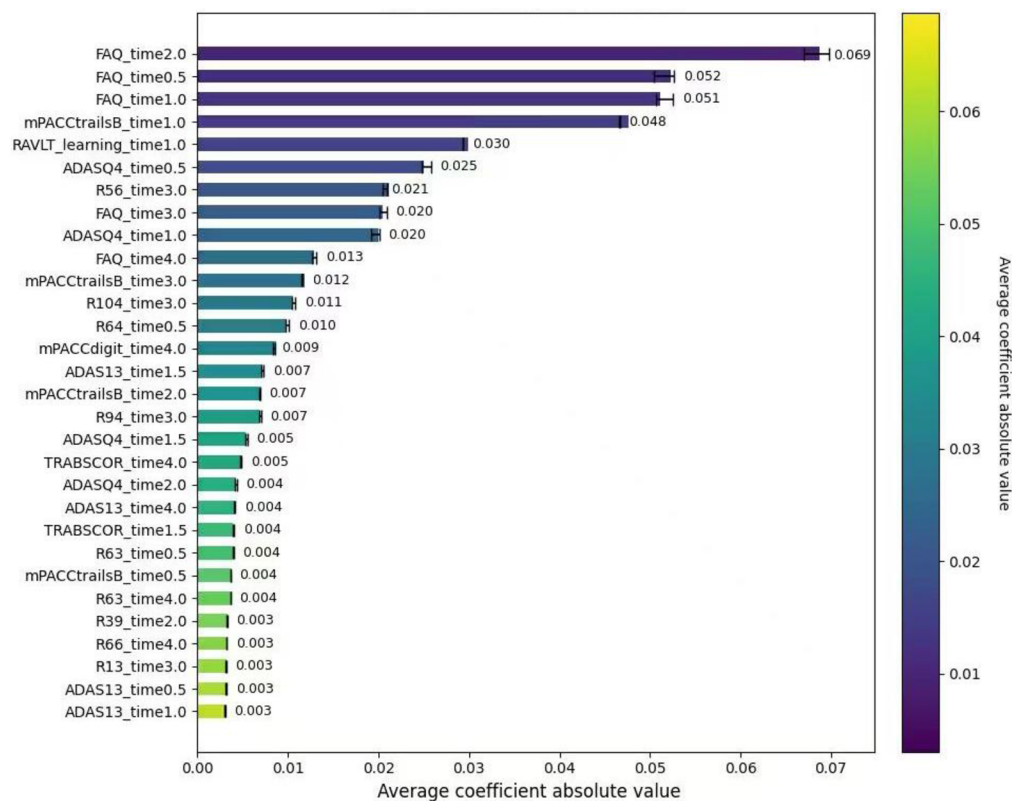
limitations of traditional statistical models, such as the need to predefine linear associations between variables and risk functions. These deep learning-based models have demonstrated superior performance compared with conventional approaches in accurately capturing the complex relationships between covariates and time-to-event outcomes [45, 47].

The results of our predictive modeling indicate that tdCoxSNN achieved robust and satisfactory performance on our multivariate longitudinal data. Compared with the

other two models, the conventional tdCox and the deep learning-based survival model Dynamic-DeepHit, tdCoxSNN consistently outperformed both. Unlike Dynamic-DeepHit, tdCoxSNN does not require the selection of discrete time intervals, thereby simplifying the handling of longitudinal data during model fitting. Notably, tdCoxSNN maintained strong predictive performance across all experimental scenarios using identical network architectures and hyper-parameters, highlighting its inherent robustness. Further tuning of these hyper-parameters



(a) Top 15 important features in group features



(b) Top 30 important features in all the longitudinal features at multiple time points

Fig. 3 Feature important results. (a) Represents top 15 important features in group features. (b) Represents top 30 important features in all the longitudinal features at multiple time points. (R56: Fusiform gyrus, R104: Thalamus, R64: Supramarginal gyrus, R63: Supramarginal gyrus, R94: Cerebellum_Crus, R39: Parahippocampal gyrus/Cuneus, R66: Angular gyrus, R13: Inferior frontal gyrus, triangular part)

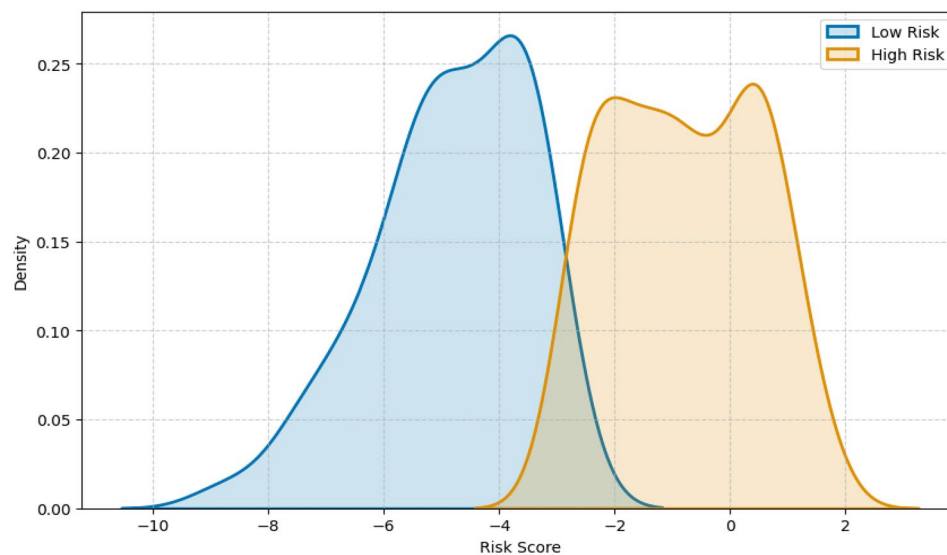


Fig. 4 Risk score distribution with gaussian mixture model grouping. The probability density graph show the estimated baseline risk score with sub-groups identified by the Gaussian mixture model

may further enhance model accuracy. The tdCoxSNN model is also computationally efficient, with the training loss typically converging within 20 epochs and requiring less than one minute per training session. Additionally, tdCoxSNN can directly integrate high-dimensional longitudinal information, such as neuroimaging data, by stacking additional neural network components. This integrated framework facilitates the development of dynamic predictive models capable of effectively leveraging complex longitudinal biomarkers, including MRI scans and omics data, within a unified neural network architecture.

We conducted a structured comparison of three distinct feature set (FS1, FS2, FS3) combinations. This systematic analysis provides empirical evidence on the incremental predictive value of each data modality, offering practical guidance for feature selection in clinical prediction models. The predictive performance was evaluated using a test set at time intervals $\Delta t = 1, 2, 3, 4$ years from the landmark time $s = 1$. The prognostic model exhibited outstanding predictive capability, attaining cAUC values of 0.9089 ± 0.01 alongside BS of 0.1651 ± 0.01 with $\Delta t = 1$ year on the test set, when all variable sets were incorporated into the tdCoxSNN model. Predictive capacity substantially diminished when excluding neuropsychological assessments, highlighting these cognitive assessments as critical predictive elements for modeling MCI-to-AD progression trajectories. We also observed that as the prediction horizon increases, the predictive performance of all three models decreases. This is primarily because for shorter horizons, this assumption is reasonably robust, as longitudinal changes are minimal, and accumulated historical

data can effectively constrain the risk score estimation. However, for longer horizons, the discrepancy between the assumed constant predictors and their true dynamic trajectories becomes amplified. Even with more accumulated historical data, the model cannot fully capture the long-term, non-stationary changes in predictors, leading to biased risk score calculations and reduced prediction accuracy.

The study conducted a comprehensive evaluation of feature importance, revealing that neuropsychological assessments played a pivotal role in predictive modeling, as evidenced by the top ten most influential variables. Among all longitudinal biomarkers, the FAQ emerged as the most important feature. The FAQ provides a straightforward and clinically applicable measure, with potential to enhance clinical utility. Previous research has shown that the FAQ is effective in distinguishing individuals with MCI from those with early-stage AD [48]. Engagement in daily activities and social interactions may confer additional benefits, potentially by promoting blood flow within the prefrontal cortex-striatum circuit [49].

From a temporal perspective, the most important features were collected within the first four years of follow-up, suggesting that early follow-up data are more informative than later measurements for predicting the progression from MCI to AD.

We also conducted a prognostic risk stratification analysis, which has substantial clinical significance for distinguishing MCI patients at high risk of progression to AD from those at low risk, who are likely to remain stable or experience slow progression. This stratification enables clinicians to implement targeted interventions and personalized management strategies more

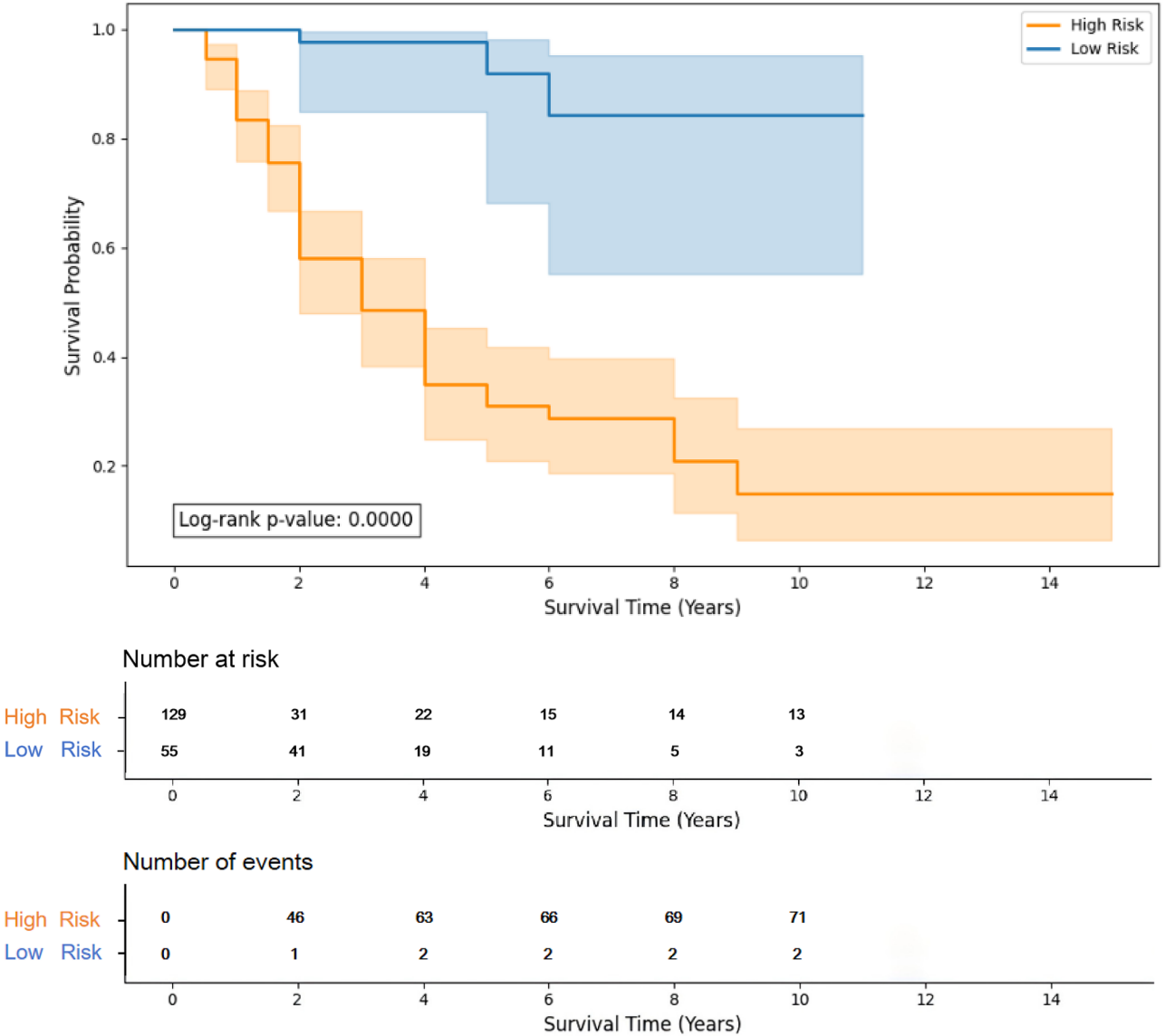


Fig. 5 The K-M estimators of the disease-free probabilities (predicted by tdCoxSNN on the test set) for high-risk and low-risk groups (log-rank test $p < 2 \times 10^{-10}$)

effectively. High-risk individuals may benefit from proactive, intensive, and tailored interventions, whereas low-risk patients can follow a more conservative management approach, focusing on maintenance of current status, preventive care, and long-term monitoring. Furthermore, risk stratification supports the prioritization of limited healthcare resources toward high-risk populations most likely to gain significant clinical benefit, thereby enhancing the efficiency and effectiveness of healthcare delivery.

Our study had several limitations. Firstly, the restricted sample size and characteristics of subjects drawn exclusively from the ADNI database. The external validation on publicly available cohorts (e.g., NACC) is not feasible in the current study, primarily due to inconsistencies in data modalities and longitudinal follow-up protocols

between these public cohorts and our research objectives for AD progression survival analysis, which would lead to incomparable and biased validation results. As a result, this study relied solely on internal validation, which may constrain the generalizability of the findings. However, The reported performance reflects internal validation within ADNI would further ensure accurate interpretation. Secondly, while utilizing longitudinal whole-brain structural neuroimaging, our study focused on region-of-interest-based volumetric measurements obtained through prior-knowledge-guided brain tissue segmentation. Future studies could employ advanced techniques like deep neural networks to capture spatial variations and subtle patterns in brain structure. Thirdly, multi-modal biomarkers can be extended to other modalities,

such as biomarkers in cerebrospinal fluid, including β -amyloid protein, total tau protein, and phosphorylated tau protein, which are key elements reflecting the pathological physiology of AD. The current investigation specifically centers on the APOE ϵ 4 genetic variant. A paramount direction for future research is the multi-modal integration of this model with CSF biomarker and multiple genetic variants to develop a more robust and clinically comprehensive prognostic tool.

Conclusion

In summary, this study developed a dynamic risk prediction model utilizing longitudinal biomarkers to estimate the risk of progression from MCI to AD. The model integrates a time-dependent Cox framework with a deep neural network, enabling accurate individualized risk prediction for MCI patients. We further evaluated the relative importance of longitudinal biomarkers and performed prognostic stratification within the MCI cohort by classifying individuals into high-risk and low-risk groups. This approach may facilitate the early identification of individuals at elevated risk and support more informed, personalized clinical decision-making.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12911-026-03387-3>.

Supplementary Material 1

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Author contributions

Rong Zhang, and Hongmei Yu conceptualized the study. Rong Zhang, Durong Chen and Yao Qin retrieved and screened the records, and extracted the data from the eligible article. Hongjuan Han and Durong Chen provided

methodological consultation. Rong Zhang drafted the manuscript and all authors revised the manuscript. All authors read and approved the final manuscript.

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Data availability

The data used in this study are available from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. Interested researchers can apply for access to the data via the official ADNI website (<https://adni.loni.usc.edu/>). All relevant materials related to the study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The Alzheimer's Disease Neuroimaging Initiative (ADNI) study was conducted in accordance with the principles of the Declaration of Helsinki. Ethical review and approval for all participating sites were obtained through their respective Institutional Review Boards (IRBs) or Research Ethics Boards (REBs), as stipulated in the ADNI study protocol. The ADNI protocols document the ethical approval requirements and are publicly available through the ADNI database (<https://adni.loni.usc.edu>). The local ethics committee of Shanxi Medical University approved the study. The ethics approval number is 2022GLL079.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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